CE205 Databases and Information Retrieval

Class 6

Objective

The purpose of this exercise is to become familiar with the Porter stemming algorithm and boolen algebra. Moreover, how to retrieve the relevant documents using a boolen query.

Problem 1

Considering the following words:

1. randomness
2. randomization

For each word, compute its stem by following the rules of the Porter stemming algorithm (see appendix) and show which rules they use.

Problem 2:

Write answers?

1 AND 1 = ___ 1 AND 0 = ___ 0 AND 1 = ___ 0 AND 0 = ___

1 OR 1 = ___ 1 OR 0 = ___ 0 OR 1 = ___ 0 OR 0 = ___

NOT (1) = ___ NOT (0) = ___

0 OR NOT ((1 AND 0) OR (NOT(0) AND 1)) = ____

Problem 3

What document(s) will be retrieved by each query from the given term-document incidence matrices?

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0
...						

Query: Brutus AND Ceaser AND (NOT Calpurnia)

Boolean Operators Applied	Term	Incidence Vector
	skeleton	(0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 1)
	ominous	(0 0 0 0 0 0 0 0 1 1 0 1 1 1 0 1 0 1)
	serenity	(1 0 0 0 0 0 0 0 1 0 1 1 1 1 0 0 0 1)

Query: ominous AND NOT (serenity OR skeleton)